

Pawara Induwara Ihalavithana

Full-stack & AI/ML Engineer
Final-year Computer Science undergraduate

CONTACT

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GITHUB
github.com/InduwaraIhalavithana

LOCATION
Sri Lanka

TECHNICAL SKILLS

Languages
Python • TypeScript • JavaScript • Java • SQL • Bash

AI / ML
XGBoost • scikit-learn • TensorFlow • PyTorch • pandas • NumPy • Isolation Forest • Autoencoders • Explainable AI

Web & Backend
React • Next.js • Tailwind CSS • FastAPI • SQLAlchemy • Alembic • PostgreSQL • REST APIs • JWT

DevOps & Cloud
Docker • nginx • GitHub Actions CI/CD • Azure VMs • Vercel

PROFILE

Final-year Computer Science and Technology undergraduate who builds complete, deployed products for real problems in Sri Lanka. I have designed and shipped three **full-stack platforms** end to end: database, REST API, React frontend, role-based dashboards, automated tests, CI/CD and cloud deployment, each in English, Sinhala and Tamil. I combine web engineering with **machine learning**, from an XGBoost crop recommender to ongoing research on AI-based counterfeit drug detection.

I became an engineer to help people: I look for problems that affect people's lives and build the software that solves them.

AT A GLANCE

3
full-stack platforms designed and shipped

243
automated tests in one project

3
languages in every app: EN · SI · TA

11
user roles designed across projects

EDUCATION

BSc (Hons) in Computer Science and Technology — 2027

Uva Wellassa University of Sri Lanka · Faculty of Applied Sciences, Badulla

Final year | Expected graduation 2027

Coursework: Machine Learning, Parallel & Distributed Computing, Computer Systems Architecture, Databases, Software Engineering

RESEARCH

Blockchain and AI-Driven Pharmaceutical Supply Chain Framework for Secure Drug Traceability and Counterfeit Detection

Final-year research | Group project | Ongoing | Poster presentation

M.R.T. Gimhan, **P.I. Ihalavithana**, N. Farook, E.M.U.W.J.B. Ekanayake · Uva Wellassa University

- **Problem:** Sri Lanka's drug supply chain (NMRA → SPC → MSD → hospitals) runs on paper records that can be forged; in 2023 about 22,500 substandard immunoglobulin vials reached state hospitals.
- **Traceability:** permissioned blockchain on **Hyperledger Fabric** with **GS1** identifiers for tamper-proof, end-to-end tracking.
- **Detection: Isolation Forest** and **Autoencoder** models flag suspicious and counterfeit transactions, with risk scores and real-time alerts stored on-chain.
- **Evaluation:** accuracy, precision, recall and F1 for the models; throughput, latency and scalability with **Hyperledger Caliper**.

MORE SKILLS

Blockchain & Tools

Hyperledger Fabric • Hyperledger Caliper • Git • LaTeX

Quality

pytest • Vitest • Testing Library • ESLint • Ruff • i18n (English / Sinhala / Tamil)

STRENGTHS

- Takes a product from idea to live deployment
- Writes tests and CI from the start
- Designs for local users: trilingual, accessible, mobile-first
- Team research collaboration

EXTRACURRICULAR

Volunteer
IEEE Student Branch, Uva Wellassa University

Volunteer
Microsoft Club, Uva Wellassa University

Member
University Tennis Team, Uva Wellassa University

LANGUAGES

Sinhala
Native

English
Professional working proficiency

INTERESTS

- AI hardware and emerging chip technologies
- Competitive tennis
- Self-development and motivational speaking

References available upon request.

PROJECTS

SmartAgri: AI-powered agriculture platform

Team of 4 · 2026

Crop advice, farm tracking and a marketplace for Sri Lankan farmers and traders

github.com/Ravindu08/SmartAgri

- **XGBoost** crop recommendation from 11 soil, climate and location inputs, with explainable results; 7-day weather for all 25 districts.
- Four roles with multi-role accounts; built the Trader module and Admin dashboard (users, reports, CSV import, harvest forecasting).
- Marketplace order lifecycle: negotiation, confirmation, simulated checkout, delivery and seller ratings.
- Docker, nginx and HTTPS on an **Azure** VM; GitHub Actions CI/CD publishes images to GitHub Container Registry.

STACK React · FastAPI · PostgreSQL · XGBoost · scikit-learn · Docker · nginx · GitHub Actions · Azure

Sharing Excess: food redistribution platform

Final-year project · 2026

Connects food donors with nearby recipients and NGOs

github.com/Induwaralhalavithana/Sharing-Excess

- Donors post surplus food; people nearby see it first and request all or part of it, with stock held until the donor responds.
- Donor, Recipient, NGO and Admin roles; NGO events, ratings, map view and guided tours.
- Live notifications over **Server-Sent Events**; installable PWA, responsive down to 320 px.
- **243 automated tests** (187 pytest, 56 Vitest), including a check that every UI string exists in all three languages.

STACK React 19 · TypeScript · TanStack Query · Zod · Leaflet · FastAPI · SQLAlchemy · PostgreSQL · Docker · GitHub Actions

ConferenceHub: conference management platform

Individual · 2026

Plan, publish and run conferences end to end

github.com/Induwaralhalavithana/Conference-Manager

- Event editor with live preview: cover image, agenda, speakers and an OpenStreetMap venue pin.
- Attendees register without an account, confirm with a 6-digit email code and get a QR ticket; a waitlist fills freed seats.
- Camera check-in desk with a live counter; analytics dashboard with a 14-day trend.
- CI runs Ruff, migration up/down checks on PostgreSQL, pytest, ESLint, TypeScript, Vitest and a production build.

STACK React 18 · TypeScript · Vite · Leaflet · FastAPI · Alembic · PostgreSQL · Gmail SMTP · Docker · GitHub Actions

Personal portfolio website

2026

github.com/Induwaralhalavithana/My-Portfolio

- **Next.js 16**, TypeScript, Tailwind CSS, shadcn/ui and Motion; prerendered, dark and light themes, English, Sinhala and Tamil; deployed on Vercel.